

Sakib Mahmud Sovon

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Research Interests

Computer Vision · Deep Learning · Multimodal Artificial Intelligence · Vision–Language Models · Natural Language Processing · Large Language Models · Accessibility Technologies · Low-Resource Language Processing · Human-Centered AI

Selected Research Contributions

- Achieved **99.56% test accuracy** on the BdSL49 benchmark with a compact 17.55 MB model — the top performer among all six architectures evaluated — published at IEEE COMPAS 2025 (IEEE Xplore).
- Delivered a complete, real-time Bengali Sign Language translation system that extends prior isolated-gesture research into continuous gesture-to-sentence generation for an underserved accessibility use case.
- Co-developed and presented NeuroPonic, a multilingual, voice-enabled LLM reasoning framework for sustainable aquaponics, at ICSASD 2026.

Education

B.Sc. in Computer Science and Engineering

Khulna University

2020 – 2025

Khulna, Bangladesh

- CGPA: 3.68 / 4.00
- **Thesis:** *Text Generation from Bengali Sign Language (BdSL) Image using Convolutional Neural Networks* — proposed and evaluated a unified YOLO-based and a two-stage YOLOv8n + CNN detection/recognition pipeline for continuous, real-time BdSL-to-text translation (see Research Experience for benchmarking details).
- **Supervisor:** Dr. Rameswar Debnath, Professor, Computer Science and Engineering Discipline, Khulna University
- **Relevant Coursework:** Machine Learning, Artificial Intelligence, Computer Vision, Digital Image Processing, Data Structures, Algorithms

Higher Secondary Certificate (HSC)

Govt. K.M.H. College

2017 – 2019

Jhenaidah, Bangladesh

- GPA: 5.00 / 5.00

Secondary School Certificate (SSC)

Kotchandpur Govt. Model Pilot Secondary School

2012 – 2017

Jhenaidah, Bangladesh

- GPA: 5.00 / 5.00

Language Proficiency

IELTS (Academic): **8.0** Overall

Publications

Conference Papers

S. P. Mistry, S. M. Sovon, M. Gain, and R. Debnath. **A Lightweighted CNN-Based Framework for Real-Time Bengali Sign Language Recognition and Translation.** *IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS 2025), Kushtia, Bangladesh, Oct. 2025.*

DOI: [10.1109/COMPAS67506.2025.11381726](https://doi.org/10.1109/COMPAS67506.2025.11381726) | [IEEE Xplore](#)

H. R. Shihab, S. M. Sovon, and M. M. Hossain. **NeuroPonic: Exploring Multilingual, Voice-Enabled LLM Reasoning for Intelligent and Sustainable Aquaponics.** *1st International Conference on Smart Agriculture for Sustainable Development (ICSASD 2026).*

S. M. Sovon and S. P. Mistry. **Text Generation from Bengali Sign Language (BdSL) Image using Convolutional Neural Networks.** *IEEE Computer Society Bangladesh Chapter (BDC) Symposium 2024, Jagannath University, Bangladesh.*

Paper details: s24.ieeecsbd.org/papers/129

Research Experience

Undergraduate Researcher — Real-Time Bengali Sign Language Translation

2024 – 2025

Computer Science and Engineering Discipline, Khulna University

Khulna, Bangladesh

- Designed the two-stage system architecture: a fine-tuned YOLOv8n detector feeding a recognition model built on MobileNetV2 (extended with GlobalAveragePooling2D, Dropout, and Dense layers) for real-time, continuous BdSL gesture-to-text translation.
- Curated and pre-processed the BdSL49 dataset (29,490 images, 49 classes): resizing, augmentation (random rotation, zoom, contrast), normalization, channel reordering, and mean-subtraction strategies tailored to each backbone.
- Ran the full benchmarking study against six baseline architectures (custom CNN, VGG16, Xception, EfficientNetV2B0, ResNet50, and MobileNetV2 without cropping) to isolate the impact of each design choice; see Selected Research Contributions for headline results.
- Designed and executed the 5-fold cross-validation and ablation protocol used to verify the model’s generalization and to attribute performance gains to specific pre-processing steps.
- Built the buffering algorithm that assembles recognized gestures into Bengali words and sentences via dedicated “space” and “end-of-sentence” gestures, including dictionary-based compound-character generation.
- Extended the thesis into the peer-reviewed IEEE COMPAS 2025 conference paper, building on an earlier paper presented at the IEEE CS Bangladesh Chapter Symposium 2024 (see Publications for full details).

Co-Researcher — NeuroPonic: Multilingual, Voice-Enabled LLM Reasoning for Aquaponics

2025 – 2026

Computer Science and Engineering Discipline, Khulna University

Khulna, Bangladesh

- Co-authored a knowledge-grounded, scenario-driven benchmark dataset for evaluating LLM decision-making in aquaponics, spanning 22 scenario batches (normal operation, water-quality

stress, biofilter failure, night-time emergencies, and post-intervention recovery) grounded in aquaculture, hydroponics, and biofilter literature and FAO guidelines.

- Contributed to a structured reasoning and evaluation protocol scoring LLM outputs across four dimensions: risk identification accuracy, action correctness, reasoning quality, and overall decision quality.
- Helped design the multilingual, voice-enabled evaluation setup, pairing each scenario with natural-language farmer queries in English and Bangla, including informal, speech-to-text-style phrasing, to test reasoning robustness across languages.
- Evaluated three frontier LLMs (GPT, Gemini, and Grok) under a strict constraint-based prompting protocol; found GPT maintained the most stable performance across English and Bangla (91–92% overall score), while Gemini and Grok showed marked degradation under Bangla and voice-transcribed conditions.
- Presented at the 1st International Conference on Smart Agriculture for Sustainable Development (ICSASD 2026).

Teaching & Mentoring

Academic Instructor

Neurone Onuraunon

June 2020 – Jul. 2024

Jhenaidah, Bangladesh

- Designed hands-on, student-centered mathematics sessions for over 5,000 students across 20 schools in the Jhenaidah district, aimed at making mathematics more approachable and engaging for secondary school students.

Teaching Assistant

Prottoy

Feb. 2023 – Jul. 2023

Remote

- Supported curriculum design and product feature ideation to improve learner engagement on the Prottoy platform.

Doubt Solver

Roots Edu

Jan. 2022 – Dec. 2022

Remote

- Provided live, forum-style academic support in Mathematics and Physics, resolving over 1,000 student doubts via chat-based Q&A.

Professional Experience

Junior Software Engineer

TechOptions

June 2025 – Present

Dhaka, Bangladesh

- Design and implement user-facing features for web-based products using React.js and FastAPI, translating product requirements into technical specifications.
- Architected concurrency-safe backend microservices (FastAPI, PostgreSQL, multi-tenant schema isolation) for LMS and ed-tech platforms, including exam, question-bank, and cognitive-skills services under high-concurrency workloads.
- Support QA testing, version control, and performance monitoring for live features.

Programmer

EDGE Project (Enhancing Digital Government and Economy)

Apr. 2026 – Jun. 2026

Dhaka, Bangladesh

- Contributed as a Programmer to iCampus, the learning management platform built for the EDGE project — a World Bank- and Government of Bangladesh-funded national digital transformation initiative under the Bangladesh Computer Council, ICT Division.

Intern Developer

Appstick

Dec. 2023 – Feb. 2024

Khulna, Bangladesh

- Contributed to the frontend architecture of a live web application, writing modular React components aligned with product specifications.

Brand Partner

Roots Edu

Jan. 2022 – Dec. 2022

Remote

- Partnered with an ed-tech platform to strengthen brand presence and support user-acquisition efforts.

Technical Skills

Machine Learning: TensorFlow, PyTorch, Keras, Scikit-learn

Computer Vision: YOLOv8, Convolutional Neural Networks, MobileNetV2, Transfer Learning, OpenCV

Programming: Python, C/C++, JavaScript (ES6+)

Research Tools: Git, NumPy, Pandas

Software Engineering: React.js, FastAPI, Node.js, Swagger, SQL, MongoDB

Other Tools: Postman, Notion, Jira (familiar)

Analytics: GA4, Metabase (introductory)

Awards & Honors

- **Champion**, University Innovation Hub Program (UIHP), Cohort-I, Khulna University — Jul. 2024 – Dec. 2024
- **1st Runner-Up, Innovation Showcasing**, SynergyX — National Symposium on Industry-Academia Collaboration for ICT Enabled Bangladesh, Khulna — Dec. 2024

Leadership & Service

President

Neurone Onuraunon

June 2020 – Jul. 2024

Jhenaidah, Bangladesh

- Led a four-year student initiative organizing mathematics outreach programs and school-level competitions under the Bangladesh Mathematical Olympiad network.
- Directed a team of volunteers, increasing regional participation and aligning initiatives with national academic objectives.
- Co-developed the organization's web platform (Node.js/Express) as a group project for the CSE

3200 Web Programming Project coursework, supporting event management and problem-solving content for members.

Convenor, Promotion Team

Dec. 2024

*SynergyX — National Symposium on Industry-Academia Collaboration for
ICT Enabled Bangladesh* *Khulna, Bangladesh*

- Led nationwide promotional strategy and managed communication campaigns for a major ICT symposium bridging academia and industry.
- Oversaw cross-university coordination, achieving record event reach and participant engagement.

Academic Team Member

Dec. 2023 – Present

Bangladesh Mathematical Olympiad *Remote / On-site*

- Organized regional and national olympiads and collaborated with local groups to expand mathematics outreach through schools, clubs, and competitions.

References

Dr. Rameswar Debnath

Professor, Computer Science and Engineering Discipline

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Dr. Manishankar Mondal

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Atanu Shome

Assistant Professor, Computer Science and Engineering Discipline

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